

CV Project Phase-1

Explainable Quantum Fusion DL for Blood Cancer Screening

Presented by:

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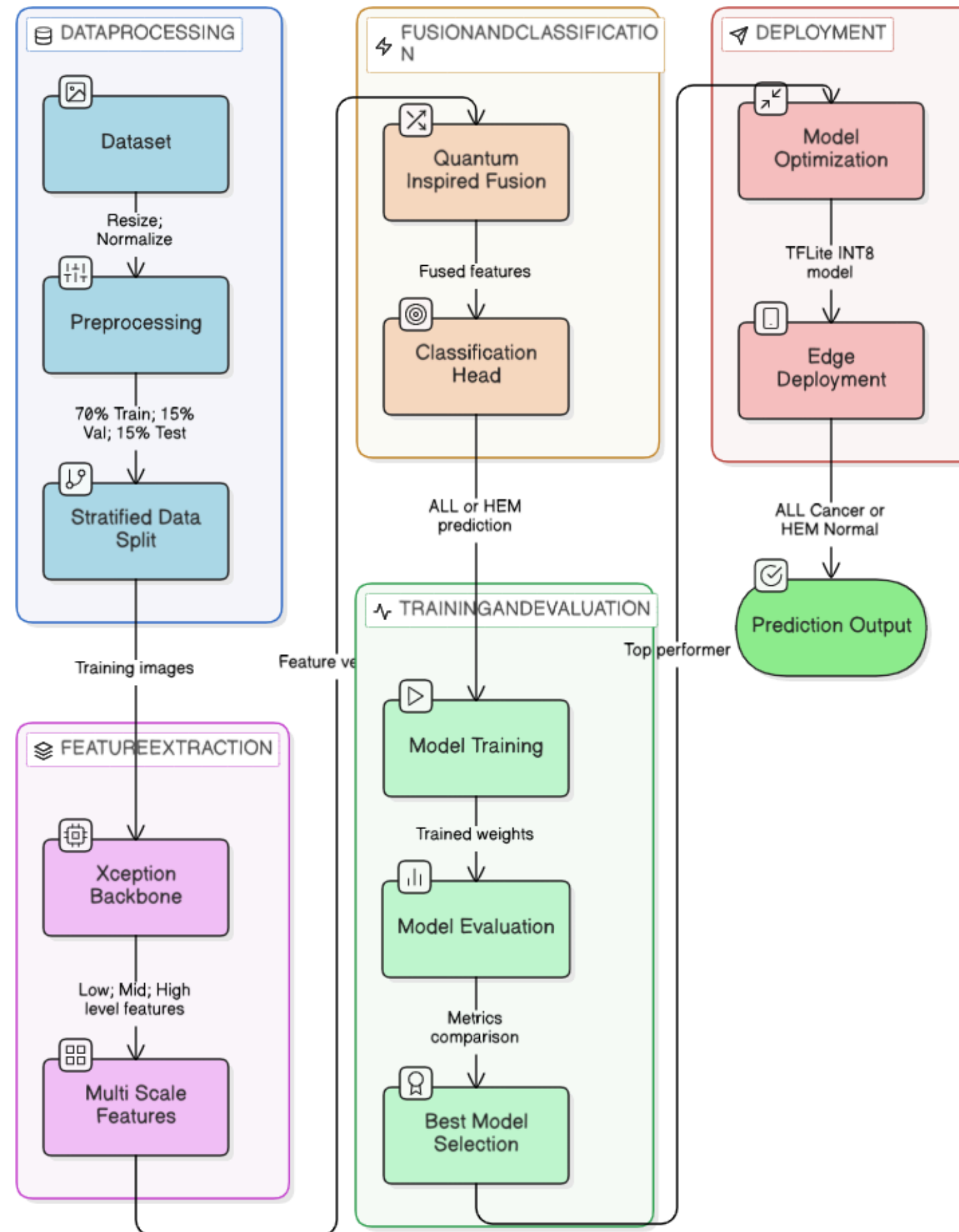
Problem Statement

- **Acute Lymphoblastic Leukemia (ALL)** is a serious blood cancer where early detection is crucial for effective treatment.
- Manual microscopic analysis of blood cells is time-consuming, expert-dependent, and prone to variability.
- Leukemic blast cells often resemble normal cells, making **accurate identification difficult**.
- The proposed work introduces a **Quantum Fusion-based Explainable Deep Learning framework** for classifying blood cell images into normal or leukemia blast cells.
- **Aggregated cell-level predictions** enable early screening of patients as cancer-positive or cancer-negative.

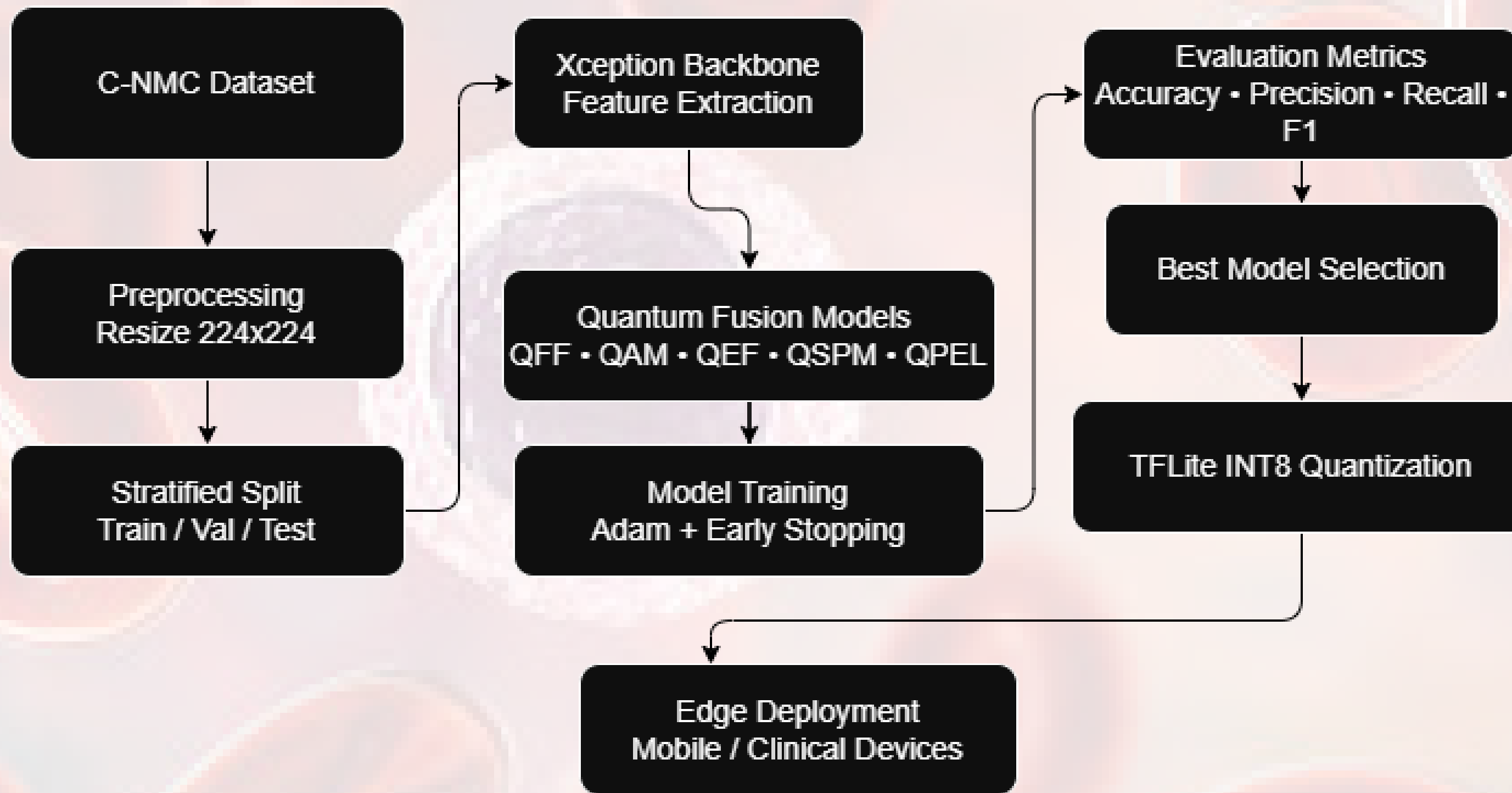
Objectives

- Develop a **deep learning model** to classify microscopic blood cell images.
- Distinguish between **leukemia blast cells (ALL)** and **normal immature cells (HEM)**.
- Use **multi-scale feature extraction** to capture detailed cellular patterns.
- Implement **quantum-inspired feature fusion techniques** to improve classification accuracy.
- Support early leukemia screening by **aggregating cell-level predictions**.

METHODOLOGY



Workflow



Techstack

Category	Tool / Component	Purpose / Description
Deep Learning Framework	TensorFlow 2.x	Core framework for model building, training, and inference
	Keras (tensorflow.keras)	High-level API for layers, models, callbacks, and optimizers
Pretrained Model	Xception	Pretrained on ImageNet, used as the CNN backbone for feature extraction
Quantum-Inspired Custom Layers (built using Keras)	QuantumFeatureFusion (QFF)	Custom layer for feature fusion
	QuantumAttention (QAM)	Custom attention mechanism layer
	QuantumEntanglement (QEF)	Custom layer simulating quantum entanglement-inspired feature interactions
	QuantumStateProjection (QSPM)	Custom layer for projecting feature states
	QuantumPhaseEncoding (QPEL)	Custom layer for encoding phase-based feature representations
Machine Learning & Evaluation	scikit-learn	Train/test split and evaluation metrics (accuracy, precision, recall, F1-score, confusion matrix, classification report)

Techstack

Data Handling	NumPy	Array operations and mathematical computations
	Pandas	Results storage and CSV/Excel export
	OpenCV (cv2)	Image handling and preprocessing
	tf.data.Dataset	Efficient data pipeline with parallel loading and prefetching
Visualization	Matplotlib	Training plots and bar charts
	Seaborn	Confusion matrix heatmaps and metric comparison bar plots
Deployment / Optimization	TFLite Converter	Converts trained Keras model to TensorFlow Lite format
	INT8 Quantization (tf.lite.Optimize.DEFAULT)	Compresses and optimizes model for edge deployment
Utilities	tqdm	Progress bars during training
	shutil	Copying best model file
	gc	Garbage collection between training runs
	time	Measuring training duration for each fusion
	os	File and directory operations

Conclusion

- The project proposes a **deep learning–based system** for automated detection of **Acute Lymphoblastic Leukemia (ALL)** from microscopic blood cell images.
- **Xception CNN with multi-scale feature extraction** helps capture important cellular patterns such as texture, shape, and structure.
- **Quantum-inspired fusion techniques** improve feature representation by combining information from different feature levels.
- The system is evaluated using **standard classification metrics** to ensure reliable performance.
- With **TensorFlow Lite INT8 optimization**, the model can be deployed on edge or mobile devices, supporting faster and accessible early screening.